

OM RAJ

Software Engineer & Web / Desktop Application Developer

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PROFESSIONAL SUMMARY

Accomplished **Software Engineer, Web Developer & Desktop Application Creator** with proven expertise in building full-stack research portals, cross-platform desktop applications published on the **Microsoft Store** (*SNIFF DOI 2.0, WASH Suite v2.0, Calculator Luxe, Clock Span*), and responsive Web UI/UX platforms. Currently associated with **IIT Ropar (Civil Engineering Dept.)** as Junior Researcher and Lead Developer for the official **AWE LAB Portal**. Specialized in JavaScript, Electron.js, Node.js, HTML5/CSS3, automated calculation engines, and federated academic data indexing engines.

</> CORE TECH STACK & SOFTWARE SKILLS

Languages & Scripts:	JavaScript (ES6+), HTML5, CSS3, JSON, Markdown, Node.js scripts.	Desktop Development:	Electron.js, Microsoft Store Packaging (MSIX/Appx), Win32 API integration.
Web & UI/UX Design:	Responsive layout math, Flexbox/Grid, Figma, GitHub Pages deployment, Google Sites.	Domain Solvers & Engines:	APHA 24th Ed & BIS 10500 calculation algorithms, CSL Citation Formatter (APA/IEEE).
Data Indexing & APIs:	Federated literature search engines (250M+ DOI index), REST API automation.	Version Control & Tools:	Git, GitHub CLI (gh.exe), VS Code, PowerShell automation, NABL PDF Generators.

PUBLISHED DESKTOP APPLICATIONS & WEB PORTALS

SNIFF DOI 2.0 — Academic Literature Discovery & Research Gap Platform

[Microsoft Store Live](#)

Developer: Om Raj | [Store Link](#)

- Engineered federated search engine indexing 250M+ scientific publications for open-access resolution.
- Developed 6-tier research gap synthesis engine and CSL reference formatter supporting APA 7th, IEEE, Harvard, BibTeX.

WASH Suite v2.0 — Environmental Chemistry Engine & Regulatory Platform

[Microsoft Store Live](#)

Developer: Om Raj (Civil Eng Dept, IIT Ropar) | [Store Link](#)

- Built Physicochemical calculation engine for APHA 24th Ed & BIS 10500 standards (pH, TDS, TSS, BOD5, COD, heavy metals).
- Implemented spectrophotometer linear regression calibration module and automated NABL ISO-17025 PDF certificate exporter.

AWE LAB Official Web Portal — IIT Ropar

[IIT Ropar Live Portal](#)

Lead Designer & Developer | [Portal Link](#)

- Designed and deployed official research laboratory portal for Dr. Indramani Dhada's team at IIT Ropar.
- Integrated laboratory member profiles, publication repositories, instrumentation showcases, and academic updates.

Calculator Luxe & Clock Span — Scientific & Chrono Desktop Tools

[Microsoft Store Live](#)

Developer: Om Raj | [Calculator Luxe](#) | [Clock Span](#)

- Created high-precision algebraic & trigonometric calculator engine with unit conversion matrix.
- Developed millisecond-accurate lab incubation timer and world clock synchronized across 400+ IANA timezones.

 PROFESSIONAL DEVELOPMENT & RESEARCH EXPERIENCE

Junior Researcher & Lead Developer

Nov 2025 – Present

Department of Civil Engineering, Indian Institute of Technology Ropar (IIT Ropar)

- Leads web portal development, research database management, and software UI/UX engineering for AWE LAB at IIT Ropar.
- Executes environmental data synthesis, wastewater compliance modeling, and spatiotemporal air quality dataset processing.

Founder & Lead Tech Creator — milliLitre

2023 – Present

milliLitre (SAH Group Packaged Water Initiative)

- Developed digital brand identity, product packaging artwork, interactive customer web tools, and marketing systems.

Freelance Web & UI/UX Developer — MARKAD LIMITED

2021 – Present

MARKAD LIMITED & Freelance Tech Professional

- Created responsive web graphics, UI components, vector branding assets, and digital communication tools.

 ACADEMIC QUALIFICATIONS

Degree / Certificate	Institution	Board / University	Year	Score
M.Sc. (Plant Pathology, Botany)	Department of Botany, Patna University	Patna University	2023 – 2025	CGPA: 7.56
B.Sc. (Botany Honours)	Patna Science College, Patna	Patna University	2020 – 2023	63.50%
Diploma in Computer Applications (DCA)	Vinita Computer Education	Vinita Computer Education	2022	64.40%
Intermediate (Science)	Shanti Niketan Sr. Sec. School	CBSE Board	2019	58.40%

 OPEN-ACCESS PORTFOLIO & SOFTWARE LINKS

- Microsoft Store Application Profile: [SNIFF DOI 2.0 on Microsoft Store](#)
- IIT Ropar AWE LAB Official Portal: <https://sites.google.com/iitrpr.ac.in/indramanidhada/awe-lab-iit-ropar>
- Personal Academic & Software Website: <https://om1904.github.io/omraj-website/>
- Google Drive Design & Software Media Repository: [Google Drive Folder](#)

 REFERENCES

1. Dr. Indramani Dhada

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2. Dr. Raj Laxmi Singh

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